Schema-Driven Document-Level Event Extraction with Coreferent Clusters: Relational Junction Scoring, Trigger-Cluster Coreference, and Balanced Positive-Negative Supervision

William Roe[1] (whr778@gmail.com), **Gemini**[2], and **Claude**[3] (noreply@anthropic.com)

[1] Project author and maintainer · [2] AI assistant (Google, Gemini) — diagnosis, architecture, experiments, and drafting · [3] AI assistant (Anthropic, Claude Opus 5) — design, implementation, and drafting

Paper 3 of the research programme stated in `RESEARCH_PROGRAM.md`. Drafted 2026-11-05; `eb22g` 5-epoch training in progress (`bcfe500c`, Epoch 1 and Epoch 2 complete, Epochs 3–5 and blind test pending).

Companion to `PAPER_0_FOUNDATION.md` (the GLiNER2 substrate and head-initialisation curve), `JOINT_IE_SCALING.md` (Paper 2 — the boundary-head data-scaling curve and constrained joint relation decoding), and `EKF_MHT_DESIGN.md` (Paper 1 — cross-document temporal state estimation, which consumes this model's document-level event records). Every experiment, positive, null, or negative, is recorded chronologically with its billed cost in `EXPERIMENT_CATALOG.md`.

Abstract

Schema-driven information extractors such as GLiNER2 take target labels as an inference-time prompt, avoiding closed ontologies, yet fail on document-level event extraction when multiple events of the same type occur in one document, when arguments lie hundreds of words from their triggers, or when an single real-world event is referred to by multiple coreferent trigger mentions across a passage. We trace this failure through four distinct architectural and optimization bottlenecks on a 4,096-token `mmBERT-base` boundary extractor and resolve each with a targeted, backwards-compatible mechanism:

1. **Instance individuation by trigger span (`event_records: true`).** Routing event supervision through the mention-structure path keys event instances by *event type*, pooling 64.2% of multi-instance document events into a single record and creating a 5× gap between relaxed (`0.5884`) and strict (`0.0991`) argument micro-F1 on `eb16-rebuild-tr`. Individuating each event instance by its trigger anchor span on the record head removes the 1-per-type pooling bottleneck without sacrificing open-schema prompting.
2. **Relational junction scoring (`record_link_mode: junction`).** Decomposing the record head's additive assignment logit `s_ijf = P_ij + R_fj` reveals that role fit `R_fj` (identical across every trigger in a group) accounts for 95% of within-row discrimination while pairwise trigger-argument compatibility `P_ij` sits below chance (per-column gold-vs-false trigger AUC `0.576`), allowing false triggers to steal 52–74% of arguments. Adding a role-gated bilinear junction term plus relative-order and log-distance geometry, trained with a contrastive per-column trigger softmax, lifts per-column junction AUC from `0.58` to `0.74–0.77` and unlocks English event argument extraction (`p5link` argument strict micro-F1 `+0.035`).
3. **Document-level coreferent trigger and argument clusters (`record_coreferent_ownership`, `TriggerLink`, `EntityCorefLink`).** A narrative article routinely mentions the same event multiple times ("Rebels *attacked* the base... The *assault* killed four"). We train every coreferent trigger mention to own its event's arguments, group coreferent and multi-occurrence argument mentions as occurrence alternatives of a single filler (`RecordFieldTarget.values`), and add a symmetric bilinear + MLP-distance `TriggerLink` scorer (pair AUC `0.802–0.843`) paired with validation-calibrated connected-component clustering (`coref_calibration.py`) and cluster-level evaluation metrics (`event_cluster` and `event_cluster_argument` strict micro-F1).
4. **Solving positive-gradient starvation under hard-negative and negative-query sampling (`eb22e` → `eb22f` → `eb22g`).** A paired role-candidate probe on `eb22e` (`3,568` gold role fillers) showed that under scheduled gold injection (`0.956`), `97.1%` of gold fillers sat in the candidate pool yet received a median assignment probability of `0.0155` because `_list_field_bce` averaged 1 positive with `K = 8` hard negatives in one denominator (equilibrium `P = 1/9 = 0.111`). Simultaneously, with gold injection off at inference, `61.7%` of gold fillers were missing from the pool or lost their trigger

because `candidate_pair_loss` with `hard_negative_keep_all_when_absent: true` kept all 768 candidates on absent queries (`~85–90%` of queries under taxonomy negative sampling) in one global denominator, diluting positive pair gradients by 7,700× (`pair_loss` pinned at `0.005–0.011`) and leaving candidate logit levels unanchored under shift-invariant listwise losses. Splitting positive and negative reductions in both `_list_field_bce` and `candidate_pair_loss` (`pair_split_reduction: true`, `hard_negative_keep_all_when_absent: false`), withholding unresolved gold roles, and eliminating double event-schema dropout lifted 64-doc overfit `@0.5` strict micro-F1 on **CASIE** from `0.3619 -> 0.5210` (`+0.1591`, peaking at `0.5984`) on `event_trigger` and `0.2506 -> 0.4319` (`+0.1813`) on `event_argument`, and on **CMNEE** from `0.6375 -> 0.9394` (`+0.3019`) on `event_trigger` and `0.5470 -> 0.8005` (`+0.2535`) on `event_argument`. On the full 263,949-document training mix (`eb22g`), Epoch 2 validation strict micro-F1 at `@0.3` reaches `event_type` **`0.9762`**, `event_trigger` **`0.2522`** (recall **`0.7434`**), `event_argument` **`0.1639`** (relaxed **`0.3992`**, fair **`0.3632`**, vs `eb22d` Epoch 2's `0.0627`), and `event_cluster_argument` **`0.1427`** (vs `eb22d` Epoch 2's `0.0579`).

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1. Introduction: Why Document-Level Events Fail in Open-Schema Extractors

Document-level event extraction asks a model to read a multi-paragraph document and emit a set of structured event records, each comprising an **event type**, one or more coreferent **trigger spans** anchoring the event in the text, and a mapping from **argument roles** to the entity or value spans that fill them. Unlike sentence-level benchmarks (ACE 2005; Walker et al., 2006) or closed-ontology taggers (OneIE; Lin et al., 2020), a generalist schema-driven extractor must satisfy four requirements simultaneously:

1. **Open schema at inference:** Event types and role inventories are supplied as prompt tokens at run time (`Schema.events({event_type: [roles]})`), not compiled into fixed classifier dimensions.
2. **Multi-instance individuation:** A news article or intelligence report often describes several distinct events of the same type (up to 16 `Attack` or `Military_Operation` instances in a single CMNEE or CASIE document).
3. **Long-range trigger-argument binding:** In document-level news (`cc_news_long`, CASIE, WikiEvents), the median trigger-to-argument distance is **446 words** (mean 1,333, 90th percentile 4,112 words; `EVENT_LINE.md`), so arguments routinely appear paragraphs away from the trigger they belong to.
4. **Cross-mention event and argument coreference:** A single real-world event is often evoked by multiple trigger mentions across a document (e.g., "*attacked*" in paragraph 1 and "*the assault*" in paragraph 4), each carrying partial role fillers that must be consolidated into one event cluster without fusing distinct events of the same type.

In earlier papers of this programme (`PAPER_0_FOUNDATION.md`, `JOINT_IE_SCALING.md`), moving from a 512-token DeBERTa-v3 span extractor to an 8,192-token `mmBERT-base` boundary extractor solved the 19-instance count ceiling and warmed the representations across 137,000 multi-task records, yet **`event_argument` strict micro-F1 remained stubbornly low (`0.069–0.118`) even when relaxed argument micro-F1 reached `0.506–0.588`**. This paper reports the series of diagnostic traces, negative results, and architectural/loss reforms (`eb16-rebuild-tr` → `eb17` → `eb18` → `eb21` → `eb22c/d/e/f/g`) that identified and solved the root causes of that gap.

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2. Bottleneck 1: Instance Individuation by Trigger Span (`event_records: true`)

2.1 The Type-Pooling Defect in Mention-Path Events (Line 2 vs Line 3)

On the incumbent boundary model `whr778/gliner2-eb16-rebuild-tr` (18,786-document blind test, evaluated at its validation-calibrated threshold `0.2`), event extraction exhibited a nearly 6× split between strict and relaxed argument F1:

Metric (`eb16-rebuild-tr` , threshold `0.2`)	Strict Micro-F1	Relaxed Micro-F1	Fair (Ortmann) Micro-F1
`event_type`	`0.8650`	`0.8650`	—
`event_trigger`	`0.5984`	`0.6210`	`0.6180`
`event_argument`	**`0.0991`**	**`0.5884`**	**`0.5737`**

Strict evaluation keys each predicted argument by `(event_type, role, span, trigger)` , whereas relaxed evaluation drops the trigger link and checks surface overlap under the event type (`EVENT_ARGUMENT_DIAGNOSIS.md`). Tracing the training and decode path revealed why: while `enable_records: true` was set in the boundary head, `event_records: true` was never enabled in the base training configs. Consequently, events were supervised through the mention-structure path, which **individuates instances by `event_type` (at most 1 instance per type per window)** and pools all triggers and arguments of the same event type into a single bag.

Across the multi-event corpora in our mixture, **64.2% of gold event instances share their event type with at least one other instance in the same document** (39.9% even share the same trigger surface string in a different sentence). Pooling them into one record per type made nearly two-thirds of gold event instances structurally inexpressible.

2.2 Anchor-Seeded Record Instances (`event_records: true`)

Enabling `event_records: true` routes event types through the boundary record head (`gliner2/models/boundary/records.py`) in `natural` (`per_query`) mode: - Role index `0` is always the **trigger** query, and roles `1..F` are the event's **argument** queries. - Each trigger candidate span `t_i` proposed for the event type seeds a distinct **event instance hypothesis** (up to `record_instance_queries = 32` per event type per window, comfortably above the worst observed document density of 16 same-type events). - Arguments are bound to specific trigger instances `t_i` via an instance-by-candidate assignment tensor `s_ijf` over trigger instance `i` , candidate span `a_j` , and role field `f` .

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3. Bottleneck 2: Relational Junction Scoring (`record_link_mode: junction`)

3.1 Why Additive Assignment Cannot Bind Arguments to Triggers

Simply turning on `event_records: true` (`eb17` , `eb18`) was not enough to bind English event arguments. In `eb18` , the record head scored trigger instance `t_i` against candidate span `a_j` for role field `r_f` additively (`records.py` `_assign_logits`):

```
s_ijf (additive) = P_ij + R_fj
                 = inst_proj(t_i) · cand_proj(a_j)   [pairwise P_ij]
                 + field_proj(r_f) · cand_proj(a_j)  [role fit R_fj, identical for all i]
```

We decomposed `s_ijf` on 16 real training batches (`456` gold arguments, `3,648` false-trigger comparisons; `JUNCTION_LAYER_SPEC.md` §1):

Diagnostic Metric	`eb18`	`p3arg-hard`
Variance across triggers `i` vs across candidates `j`	`18.7` vs `44.1`	`4.3` vs `12.8`
Per-column Junction AUC (gold trigger vs false trigger on same gold argument `a_j`)	**`0.631`**	**`0.576`**
Per-row Argument AUC (`R_fj` only / `P_ij` only / `P_ij + R_fj`)	`0.947` / **`0.405`** / `0.948`	`0.953` / **`0.371`** / `0.954`

Because `R_fj` does not depend on trigger `i` , **the additive score is 95% role fit with no relational join**. A candidate span that looks like a plausible `Victim` or `Attacker` receives virtually the same high score on *every* trigger row in the document, so false or sibling triggers steal 52–74% of arguments, and pushing false trigger rows down in training depresses the shared role term `R_fj` and destroys recall.

3.2 Gated Bilinear + Geometry Junction Layer and Contrastive Column Loss

In relational terms, `Trigger` (`t_i`) and `Argument` (`a_j`) are many-to-many entities that require an explicit join table keyed by `(i, j, f)`. Under `record_link_mode: junction`, we augment the assignment logit with a **role-gated bilinear link** and a **relative geometry** term:

```
s_ijf (junction) = P_ij + R_fj
                 + <(U t_i) ⊙ (V a_j) ⊙ σ(G q_f), 1> / sqrt(d)   [gated bilinear link]
                 + w_f · e_geom(i, j)                            [order + log distance]
```

where `e_geom(i, j)` encodes relative span order and log-bucketed token distance between trigger span `t_i` and argument span `a_j`, and `V, w_f` are zero-initialized so warm starts are bit-identical at step 0.

Crucially, rather than training false trigger rows with absolute negative BCE targets (which depresses `R_fj`), we supervise the junction **contrastively down each gold argument's column**: for every gold argument column `j` of role `f`, a listwise cross-entropy loss (`record_link_column_weight: 0.3`) maximizes the log-probability mass assigned to the gold trigger instance(s) `G_j` against all competing trigger instances in that column.

In a controlled A/B (`p5link`, 2026-10-04), enabling the junction layer and column loss lifted per-column junction AUC from `0.58` to **`0.74–0.77`** and improved `event_argument` strict micro-F1 by **`+0.035`** (with `eb22f` and `eb22g` maintaining cumulative junction AUC of **`0.728–0.760`** over 200,000+ natural groups).

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4. Bottleneck 3: Document-Level Coreferent Event and Argument Clusters

4.1 Multi-Mention Event Ownership (`record_coreferent_ownership: true`)

Real-world news articles frequently refer to the same event through multiple coreferent triggers across a document (`cc_news_events_sonnet55_v2` contains 9,528 positive coreferent trigger pairs across 3,760 multi-mention events). Previously, when an event record listed multiple coreferent triggers `triggers = [t_1, t_2, ...]`, `compute_group_loss` seeded a gold instance only on `t_1` (`cols[0]`): `t_2` was protected from negative supervision but received **zero argument training**, and at decode time both `t_1` and `t_2` could fire as separate duplicate events that split the event's arguments between them (`COREFERENT_OWNERSHIP_SPEC.md`).

Under `record_coreferent_ownership: true`, **every coreferent trigger mention `t_m` in cluster `G` seeds a gold instance row and owns the event's argument targets**, with the record's field loss normalized by `1 / |G|` so multi-mention events do not overweight the gradient.

4.2 Learned Trigger Coreference (`TriggerLink`) and Validation-Calibrated Clustering

Because `record_coreferent_ownership: true` trains every trigger mention of an event to emit its arguments, decode must consolidate mentions of the *same* event into a single cluster without merging distinct same-type events.

A first heuristic — merging same-type decoded instances that shared a high-scoring argument (`3b-i`) — **traced negative**: on 15 multi-trigger validation documents, **0 of 6 merged events matched a gold cluster**, because argument-score cosine is dominated by the shared role term `R_fj` (pairwise AUC **`0.468`**, below chance; `COREFERENT_LINK_SPEC.md` §1). By contrast, the cosine of `inst_proj` trigger states already carried an untrained coreference AUC of **`0.709`**.

We therefore added `TriggerLink` (`record_coref_link: true`), which scores trigger instance pairs `(t_i, t_j)` within an event type via a symmetric bilinear projection plus a 64-width MLP over pair geometry (`coref_link_distance: mlp`, `coref_link_distance_dim: 64`, taking `log(1 + Δ_tok)`, same-sentence indicator, and surface-match indicator):

```
s_ij (coref) = <W t_i, W t_j> / sqrt(d) + MLP_geom(g(i, j))
```

Why an MLP for distance? In the `eb22c` score-level autopsy (`EB22C_SCORE_LEVEL_AUTOPSY.md` §10), a linear scalar distance weight accounted for 66–68% of the coreference logit; replacing it with a zero-initialized 64-unit MLP was the **only** `eb22c` architectural arm that cleared the three-control noise floor on both validation corpora (catalog 2026-10-23). Across `eb22f` and `eb22g`, `TriggerLink` achieves **pair AUC `0.802–0.843`** over `19.8M` mention pairs.

At the end of training, `gliner2/training/coref_calibration.py` sweeps the merge threshold `τ_coref` on validation (`cc_news_events_sonnet55_v2`), gated by: 1. At least 20 merges performed, 2. Pairwise merge precision ≥ 0.80, 3. Distinct-event fusion rate < 0.05, and 4. Non-decreasing cluster F1 on both the calibration set and single-trigger controls (`cmnee`, `maven`).

If no threshold clears all four gates, `record_merge_coreferent` remains `"off"` in `config.json` rather than shipping a destructive merge.

4.3 Argument Coreference and Multi-Occurrence Target Compilation

Document-level benchmarks (WikiEvents, RAMS, ACE, and our LLM-annotated news corpora) score an extracted argument as correct if it matches *any* coreferential mention of the gold entity, or if the canonical entity surface appears multiple times in the document. Our audit (`2026-11-04`, commit `0607b29`) uncovered a critical target-compilation bug in how both coreferent argument mentions (`expand_argument_coreferents` in `gliner2/processing/argument_coref.py`) and multi-occurrence string-gold arguments (`_build_outputs` in `gliner2/processor.py`) were handed to the record head:

- **Before (`eb22d` – `eb22f`):** When `expand_argument_coreferents` appended a coreferent mention `"he"` to `"John Smith"`, or when a string-gold argument `"malware"` appeared at 4 token spans in a CASIE document, `_build_record_field_target` (`boundary_preprocessing.py`) compiled each occurrence as a **separate required filler** in `RecordFieldTarget.values` (`len(ft.values) = K`). During training, `_list_field_bce` then forced the model to push *all* `K` *columns simultaneously toward* `1.0` for a single role, and treated any occurrence that fell outside the candidate pool as an unresolved missing role!
- **After (`0607b29` / `eb22g`):** `expand_argument_coreferents` tags coreferent expansions with `_coref_group`, and `_build_outputs` wraps each listed argument's document matches as one occurrence group, so `_build_record_field_target` compiles **one** value entry whose tuple holds all acceptable span alternatives: `ft.values = (((span_1), (span_2), ...),)` (`len(ft.values) = 1`). Resolving *any* coreferent or repeated mention in the candidate pool satisfies the role filler.

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5. Bottleneck 4: Positive-Gradient Starvation in Role Assignment and Span Proposal

5.1 The `eb22e` Paired Role-Candidate Probe: Separating Assignment from Proposal

After zero-initializing the post-`eb21` span residuals (`span_residual_zero_init: true`, `enable_rotary_endpoints: false` in `eb22e`), Epoch 1 validation `event_argument` strict micro-F1 sat at `0.0755` (at swept threshold `0.1`), and blind-test evaluation on WikiEvents at flat threshold `0.5` scored `0.0000` on every event head (`EXPERIMENT_CATALOG.md` 2026-11-02).

To separate **role-assignment failure** from **span-proposal failure** without confounding from random schema dropout, we ran a **paired-collate probe** (`probe_role_candidates.py`, `2026-11-03`) on `eb22e:checkpoint-epoch-1` across `3,568` gold role fillers (`20` val docs each of `casie`, `cmnee`, `duee` + all `19` `wikievents.dev`), collating each batch **once** and running two forwards — first at Epoch 1's scheduled gold injection (`0.9559`), then with gold injection off (`0.0`, the inference regime):

Regime (`n = 3,568` identical gold fillers)	Candidate in Pool	Not in Pool	Trigger Not Proposed	Median Assignment `P`	Mean Assignment `P`	Share `P ≥ 0.1`	Share `P ≥ 0.5`
Scheduled injection (`0.9559`)	**`97.1%`** (`3,466`)	`2.3%`	`0.6%`	**`0.0155`**	`0.0674`	`18.0%`	**`2.1%`**
Inference (`0.0` injection)	**`38.3%`** (`1,366`)	**`28.8%`**	**`32.9%`**	**`0.0280`**	`0.0994`	`25.2%`	**`4.4%`**

Per-corpus candidate reachability at `0.0` injection exposed a sharp split: - **Chinese sentence/short-doc corpora:** `duee` **`100.0%`** candidate (`median P = 0.0684`), `cmnee` **`96.4%`** candidate (`2.4%` not in pool, `1.2%` trigger missing, `median P = 0.0452`). - **English multi-paragraph corpora:** `casie` **`34.8%`** candidate (`29.5%` not in pool, `35.7%` trigger missing, `median P = 0.0048`), `wikievents` **`30.0%`** candidate (`32.9%` not in pool, `37.1%` trigger missing, `median P = 0.0352`).

This single table proved that **two independent starvations** were operating at the same time: 1. **Even when `97.1%` of gold fillers were present in the candidate pool, their median assignment probability was `0.0155`** (only `2.1%` reached `0.5`). 2. **When gold injection was turned off at inference, `61.7%` of gold fillers were not candidates at all** (`28.8%` span not in pool, `32.9%` trigger not proposed).

5.2 Solving Role-Assignment Starvation (`overfit-events-eb22e` → `eb22f`)

Tracing `_list_field_bce` (`gliner2/models/boundary/records.py`) and `_format_field` (`gliner2/inference/engine.py`) identified four defects depressing role assignment scores:

1. **Combined `(1 + K)` denominator in `_list_field_bce`:** With `record_role_hard_negatives: 8`, `_list_field_bce` concatenated `G` gold columns (`target = 1`) and `K = 8` hardest negative columns (`target = 0`) and took a single `.mean()`. For a 1-filler role (`G = 1, K = 8`), the positive gradient weight was `1/9` while the negative weight was `8/9`, placing the zero-gradient equilibrium at `P = 1 / (1 + K) = 0.111` whenever hard negatives shared features with the gold filler. **Fix:** Split the reduction into `pos_loss.mean() + neg_loss.mean()`, giving the positive set and hard-negative set equal total weight regardless of `G` and `K`.
2. **Training missed gold roles as empty (`target = 0`):** When a gold role had `len(ft.values) > 0` but none of its spans entered the candidate pool (`gold_cols = []`), `_list_field_bce` trained the hardest candidate columns toward `0.0`, teaching the model to suppress valid role fillers whenever the proposer missed the exact gold boundaries, and masking out partially overlapping spans on partial matches. **Fix:** Withhold unresolved non-empty roles (`0.0` loss) in both sparse (`compute_group_loss`) and dense (`compute_dense_batch_loss` / `build_dense_record_matching_cost`) paths, and mask candidates that overlap unresolved gold spans on partial matches (`_unresolved_overlap_cols`).
3. **Anchor self-row in field loss (`record_drop_anchor_field_loss: true`):** In `natural` mode, `field_idx = 0` is the trigger anchor that seeded the instance, yet its row was included in `field_loss_sum / len(spec.fields)`. Dropping the anchor self-row focuses `record_field_loss` exclusively on argument roles.
4. **Double-gating at format time (`engine.py`):** `decode_group` already gated role fillers by `assign_prob = sigmoid(assign_logit / T) >= threshold`, and `engine.py` then overwrote the filler's confidence with `min(assign_prob, span_score)` (where `span_score` was the uncalibrated raw pair score), causing `_format_field` to discard valid arguments whose assignment probability cleared the threshold. **Fix:** Store the assignment probability as the field confidence (`span_score` preserved as `pair_confidence` diagnostic).

Experimental validation (`overfit-events-eb22e`, 1x A10, 64 docs × 400 steps on `eb22e:checkpoint-epoch-1`):

Corpus (64 docs, Step 400, injection off at eval)	Arm	Median Gold Role P	Share P ≥ 0.5	event_argument Strict Micro-F1 @0.5	event_argument Strict Micro-F1 @0.3	event_argument Strict Micro-F1 @0.1
casie_off	Control (legacy role loss)	0.0316	13.5%	0.1389	0.1779	0.2437
casie_off	**Treatment (eb22f role loss + gate)**	**0.5545**	**53.4%**	**0.1938** (+0.0549)	**0.2407** (+0.0628)	**0.2567** (+0.0130)
cmnee_ner_off	Control (legacy role loss)	0.5078	50.4%	0.3119	0.4000	0.5238
cmnee_ner_off	**Treatment (eb22f role loss + gate)**	**0.9196**	**82.0%**	**0.5778** (+0.2659)	**0.5882** (+0.1882)	**0.5872** (+0.0634)

Even at **Step 0** (frozen checkpoint, isolating the `_format_field` confidence fix alone), `cmnee_ner_off` argument strict micro-F1 `@0.5` jumped from **`0.105 -> 0.190`** (`+0.085`). By Step 400, median gold role assignment probability rose **`0.0316 -> 0.5545`** on CASIE and **`0.5078 -> 0.9196`** on CMNEE (`+0.266` argument F1 `@0.5`).

However, when we trained a full Epoch 1 of the 263,949-document mix with these fixes (`eb22f-epoch`, 4x H100 SXM5, `step 16,585`), validation strict micro-F1 at `@0.1` improved substantially on **`entity`** (**`0.4262 -> 0.4679`**, **`+0.0417`**) and **`relation`** (**`0.1326 -> 0.1665`**, **`+0.0339`**), while **`event_argument` remained at `0.0767`** (vs **`eb22e`**'s **`0.0755`**) and `event_trigger` remained at `0.1619` (vs `0.1608`). That confirmed the second half of the paired probe's prediction: during Epoch 1, gold injection stays at `1.0 -> 0.96`, and without fixing **why the proposer fails to propose 61.7% of gold spans when injection is off**, role-loss improvements cannot surface at inference.

5.3 Solving Candidate Proposal Starvation (`overfit-events-eb22g` → `eb22g`)

Why was the boundary proposer failing to propose English event triggers and arguments — and why had `eb22c` exhibited score-level swings where `pointwise pair_loss` sat at `~0.005` against `listwise ~3.5` (`EB22C_SCORE_LEVEL_AUTOPSY.md` §4)?

Tracing `candidate_pair_loss` (`gliner2/models/boundary/losses.py`) and `processor.py` revealed the exact mechanism:

1. **Absent-query negative dominance in `candidate_pair_loss`:** Under taxonomy negative sampling (`negative_labels_per_dim: {entities: 20, events: 20}` + `gold_full_roles: true`), 85–90% of queries in a training batch have **zero positive spans** (`pos_count == 0`). In `eb22c` – `eb22f`, `hard_negative_keep_all_when_absent: true` kept **all 768 candidates** as negatives on every absent query, and `candidate_pair_loss` divided the sum of all positive and negative BCE losses by `total_mask.sum()`. - **Traced on 1 positive span at logit `-5.0` + 10 absent queries × 768 candidates at logit `-3.0`:** `total_mask.sum()` is `1 + 20 + 7,680 = 7,701`. The positive span's BCE loss (`5.0067`) is divided by `7,701`, shrinking its gradient on the positive logit from `-0.9933` (`= P - 1`) to **`-0.000129`** (7,700× smaller!) and shrinking `pair_loss` to **`0.0492`**. - Meanwhile, the listwise losses (`proposal_loss` and `rerank_listwise_loss`) are softmaxes over candidates within positive queries only — which are **shift-invariant** (`softmax(s + c) = softmax(s)`). With the only level-anchoring positive term crushed by 7,700× and absent queries pushing all candidates down, the overall logit level drifted below the `-0.85` (`threshold = 0.3`) decode gate! - **Fix (`eb22g`):** Added `pair_split_reduction: true` (`split_positive_negative=True` in `candidate_pair_loss`), which averages positive BCE over `pos_mask.sum()` and negative BCE over `neg_mask.sum()` (`0.5 * (pos_mean + neg_mean)` when both exist), and set `hard_negative_keep_all_when_absent: false` so absent queries sample at most `minimum_hard_negatives: 16` hardest false positives instead of all 768 candidates. On the same traced batch, `pair_loss` moves **`0.0492 -> 5.0553`** and the positive logit gradient restores to **`-0.993307`**.
2. **Double event-schema dropout (`remove_events_prob: 0.0`):** Even after `NegativePoolSampler` constructed the training menu, `SchemaDropout` in `_process_events` randomly dropped `20%` of event types (`remove_events_prob:`

`0.2`) — discarding a fifth of gold event supervision on every step, and even firing inside the inference dict-schema branch if `is_training` remained set! We removed `remove_events_prob` from the inference branch and set `remove_events_prob: 0.0` in `eb22g.yaml`, increasing supervised gold role fillers per step by ~25%.

3. **Evaluation hygiene (`train.py`, `eval_metrics.py`):** Fixed `_gold_event_trigger_set`, `_gold_event_argument_set`, and `_gold_oneie_sets` to unwrap `{text, start, end}` offset dicts (`casie_off` perfect-prediction F1 `0.0 -> 1.0`), and fixed standalone `evaluate_config` to exclude `data.train_only` corpora and run `check_and_clean`.

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6. Experimental Results

6.1 Controlled 64-Document Overfit A/B: `overfit-events-eb22g` (`eb22f` vs `eb22g`)

Before committing to a 5-epoch training run, we ran a controlled 4-arm experiment (`overfit-events-eb22g`, `1x A100 40 GB SXM4`, instance `37b84a7e`, commit `a0d8e11`) starting from the exact same `whr778/gliner2-eb22f-epoch:checkpoint-epoch-1` weights across 64 training documents of **`casie_off`** (English multi-paragraph cybersecurity news: 609 gold triggers, 1,518 gold arguments) and **`cmnee_ner_off`** (Chinese military news: 122 gold triggers, 399 gold arguments) for 400 steps (`batch_size = 4`, `grad_accum = 1`). Both arms included the occurrence/coref target grouping (`0607b29`); **Control** used the `eb22f` settings (`pair_split_reduction=False`, `hard_negative_keep_all_when_absent=True`, `remove_events_prob=0.2`) and **Treatment** used the `eb22g` settings (`pair_split_reduction=True`, `hard_negative_keep_all_when_absent=False`, `remove_events_prob=0.0`). All evaluations ran with gold injection **off** (`is_training=False`):

Table 1: `casie_off` (64 records: 609 triggers, 1,518 arguments — Strict Micro-F1)

Metric	Prior `eb22e` Treatment (Step 400)	Step 0 (`eb22f` ckpt)	Step 400 Control (`eb22f`)	Step 400 Treatment (`eb22g`)	Delta (`eb22g` − `eb22f`)
`event_type` strict micro-F1 `@0.5`	`0.8618`	`0.0000`	`0.9384`	**`1.0000`**	**`+0.0616`**
`event_type` strict micro-F1 `@0.3`	`0.9163`	`0.0000`	`0.9533`	**`1.0000`**	**`+0.0467`**
`event_type` strict micro-F1 `@0.1`	`0.9609`	`0.0177`	`0.9581`	**`1.0000`**	**`+0.0419`**
`event_trigger` strict micro-F1 `@0.5`	`0.3889`	`0.0000`	`0.3619`	**`0.5210`** *(step 350: **`0.5984`**)*	**`+0.1591`**
`event_trigger` strict micro-F1 `@0.3`	`0.3824`	`0.0000`	`0.3877`	**`0.4554`** *(step 350: **`0.5506`**)*	**`+0.0677`**
`event_trigger` strict micro-F1 `@0.1`	`0.3712`	`0.0037`	`0.3982`	`0.3731` *(step 350: **`0.4741`**)*	`−0.0251`
`event_argument` strict micro-F1 `@0.5`	`0.1938`	`0.0000`	`0.2506`	**`0.4319`**	**`+0.1813`**
`event_argument` strict micro-F1 `@0.3`	`0.2407`	`0.0000`	`0.2871`	**`0.4108`**	**`+0.1237`**
`event_argument` strict micro-F1 `@0.1`	`0.2567`	`0.0025`	`0.2857`	**`0.3303`** *(step 350: **`0.4110`**)*	**`+0.0446`**
`pair_loss` (step 50 → step 400)	—	`0.0000`	`0.0109` → `0.0113` *(flat)*	**`1.6638` → `0.4118`**	actively trains
`record_field_loss` (step 400)	`0.3316`	`0.0000`	`0.3093`	**`0.2793`**	**`−0.0300`**
`gold_role_n` (steps 350–400)	`3,967`	—	`4,149`	**`5,119`**	**`+970 (+23.4%)`**
`gold_role_p_median` (step 400)	`0.5545`	—	`0.6316`	**`0.7058`**	**`+0.0742`**

Table 2: `cmnee_ner_off` (64 records: 122 triggers, 399 arguments — Strict Micro-F1)

Metric	Prior eb22e Treatment (Step 400)	Step 0 (eb22f ckpt)	Step 400 Control (eb22f)	Step 400 Treatment (eb22g)	Delta (eb22g − eb22f)
event_type strict micro-F1 @0.5	0.8584	0.1798	0.7538	**0.9811**	**+0.2273**
event_type strict micro-F1 @0.3	0.8780	0.2198	0.8000	**0.9938**	**+0.1938**
event_type strict micro-F1 @0.1	0.9032	0.4118	0.8175	**0.9938**	**+0.1763**
event_trigger strict micro-F1 @0.5	0.6512	0.1308	0.6375	**0.9394**	**+0.3019**
event_trigger strict micro-F1 @0.3	0.6909	0.1622	0.6905	**0.9100**	**+0.2195**
event_trigger strict micro-F1 @0.1	0.7200	0.2927	0.7119	**0.8276** *(step 250: **0.8829**)*	**+0.1157**
event_argument strict micro-F1 @0.5	0.5778	0.0964	0.5470	**0.8005**	**+0.2535**
event_argument strict micro-F1 @0.3	0.5882	0.1137	0.5968	**0.7543** *(step 300: **0.8099**)*	**+0.1575**
event_argument strict micro-F1 @0.1	0.5859	0.1367	0.5859	**0.6780** *(step 300: **0.7786**)*	**+0.0921**
pair_loss (step 50 → step 400)	—	0.0000	0.0052 -> 0.0060 *(flat)*	**1.1424 -> 0.2650**	actively trains
record_field_loss (step 400)	0.5625	0.0000	0.5092	**0.3545**	**−0.1547**
gold_role_n (steps 350–400)	1,100	—	952	**1,260**	**+308 (+32.4%)**
gold_role_p_median (step 400)	0.9196	—	0.9763	**0.9797**	**+0.0034**

Three findings stand out from Tables 1 and 2: 1. **Occurrence/coreferent target grouping alone lifted casie_off @0.5 argument F1 from 0.1938 -> 0.2506 (+0.0568)** between overfit-events-eb22e 's treatment and overfit-events-eb22g 's control. 2. **Restoring positive pair_loss broke the ~0.25 CASIE proposal ceiling**, lifting casie_off @0.5 trigger strict micro-F1 to **0.5210** (0.5984 at step 350) and argument strict micro-F1 to **0.4319** (+0.1813 over control, +0.2381 over overfit-events-eb22e). 3. **Threshold inversion disappeared:** Because positive pair logits are now anchored above 0.0 , @0.5 is the strongest operating point on both CASIE (0.4319 vs 0.3303 at @0.1) and CMNEE (0.8005 vs 0.6780 at @0.1).

6.2 Full-Scale Training Progression: eb22d vs eb22e vs eb22f vs eb22g (Epochs 1–2, Live)

We trained eb22g from scratch (from_encoder on jhu-clsp/mmBERT-base , seed 42, 263,949 cleaned training records → 265,369 4,096-token chunks, 19,286 validation records, 20,602 blind-test records, 82,930 steps across 5 epochs) on 1x A100 40 GB SXM4 (bcfe500c). At step 20, boundary_head.pair_scorer registered a gradient norm of 34.8 (4th largest in the model), compared to near-zero in eb22d – eb22f .

Table 3 compares the 1,500-document validation draw at the end of **Epoch 1 (step 16,586)** and **Epoch 2 (step 33,172)** across the eb22 lineage (single-run noise floor **±0.02**, relations **±0.041**):

Table 3: Full-Mix Validation Progression Across the `eb22` Series

Metric (Validation, Strict Micro-F1 unless noted)	`eb22d` Ep 1 (`@0.1` / `@0.3`)	`eb22e` Ep 1 (`@0.1`)	`eb22f` Ep 1 (`@0.1`)	`eb22g` Ep 1 (`@0.3`)	`eb22d` Ep 2 (`@0.3` full val)	`eb22g` Ep 2 (`@0.3`)	**Delta (`eb22g` Ep 2 − `eb22d` Ep 2)**
Selected Swept Threshold	`0.1` *(0.3)*	`0.1`	`0.1`	**`0.3`**	`0.3`	**`0.3`**	Matched at `0.3`
`event_argument` strict micro-F1	`0.1123` *(0.0058)*	`0.0755`	`0.0767`	**`0.0974`**	`0.0627`	**`0.1639`**	**`+0.1012`** *(2.61×)*
`event_argument` strict Precision / Recall	—	`0.237` / `0.045`	`0.142` / `0.053`	`0.063` / **`0.221`**	`0.296` / `0.035`	`0.115` / **`0.284`**	**Recall `8.1×` higher** (`28.4%` vs `3.5%`)
`event_argument` relaxed micro-F1	—	`0.1021`	`0.0985`	**`0.2620`**	—	**`0.3992`** *(R= `0.6448`)*	**`+0.3007` vs `eb22f` Ep 1**
`event_argument` fair (Ortmann) micro-F1	—	`0.0989`	`0.1026`	**`0.2191`**	—	**`0.3632`**	**`+0.2606` vs `eb22f` Ep 1**
`event_type` strict micro-F1	`0.6478`	`0.5988`	`0.3848`	**`0.9681`**	—	**`0.9762`**	**`+0.5914` vs `eb22f` Ep 1**
`event_trigger` strict micro-F1	`0.1608` *(0.1334)*	`0.1608`	`0.1619`	**`0.1610`** *(R= `0.573`)*	`0.3645` *(R= `0.316`)*	**`0.2522`** *(R= **`0.743`**, rel R= **`0.790`**)*	Trigger recall **`74.3%`** vs `31.6%`
`event` (combined) strict micro-F1	—	—	`0.2109`	**`0.2543`**	—	**`0.3505`** *(rel **`0.3988`**)*	**`+0.1396` vs `eb22f` Ep 1**
`event_cluster` strict micro-F1	—	—	`0.1729`	**`0.1694`**	`0.3660`	**`0.2584`**	**`+0.0890` Ep 1 → Ep 2**
`event_cluster_argument` strict micro-F1	—	—	`0.0635`	**`0.0821`**	`0.0579`	**`0.1427`**	**`+0.0848`** *(2.46× `eb22d` Ep 2)*
`classification` strict micro-F1	`0.4921`	`0.4802`	`0.4645`	`0.4574`	—	**`0.5442`**	**`+0.0797` vs `eb22f` Ep 1**
`entity` strict micro-F1	`0.3617` *(@0.1)*	`0.4262` *(@0.1)*	`0.4679` *(@0.1)*	`0.2867` *(@0.3, R= `0.512`)*	—	`0.2445` *(@0.3, R= `0.540`–`>0.678`)*	Evaluated at `@0.3` (per-head sweep at Ep 5)
`relation` strict micro-F1	`0.1340` *(@0.1)*	`0.1326` *(@0.1)*	`0.1665` *(@0.1)*	`0.0448` *(@0.3, P= `0.309`)*	—	`0.0562` *(@0.3, P= `0.659`–`>0.835`)*	Evaluated at `@0.3`
`structure` strict micro-F1	`0.1691` *(@0.1)*	`0.2231` *(@0.1)*	`0.1501` *(@0.1)*	`0.0648` *(@0.3)*	—	`0.0847` *(@0.3, R= `0.287`)*	Evaluated at `@0.3`
`head_min` (`eval_overall_strict_head_min_f1`)	—	`0.0755` *(@0.1)*	`0.0767` *(@0.1)*	`0.0448` *(@0.3)*	—	**`0.0562`** *(@0.3)*	Rising monotonically at `@0.3`

(Epochs 3, 4, and 5, post-training `threshold_sweep.json` + per-head thresholds, `coref_calibration.json`, and the 20,602-record blind test `test_metrics.json` will be appended here upon completion of `bcfe500c`.)

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7. Downstream Connection: Providing the Event Substrate for EKF/MHT

In `RESEARCH_PROGRAM.md` and `EKF_MHT_DESIGN.md`, the cross-document temporal tracker (Paper 1) requires an upstream extractor that converts each news article into structured event observations `(event_type, trigger_cluster, role -> value)` before the Viterbi association and Kalman filter stages run. Earlier EKF front-end runs (`EVENT_LINE.md` §4a) were blocked because: 1. Mention-path event extraction pooled all events of the same type in an article into one record, blending casualties or locations from separate incidents; 2. Without the junction layer, false triggers stole arguments from true triggers; and 3. Without `TriggerLink` coreference and positive-gradient proposal supervision, multi-mention events either fragmented into duplicate observations or failed to propose spans on long English news articles.

With `eb22g` uniting trigger-anchored record instances, relational junction scoring, coreferent trigger/argument clustering, and balanced positive-negative proposal and role supervision, the event extraction front-end now meets the structural contract required to resume the EKF/MHT multi-event tracking experiments once Epoch 5 completes.

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8. Reproducibility & Code Map

Mechanism	Primary Implementation	Config Flag (`base/eb22g.yaml`)
Trigger-seeded event records	`gliner2/models/boundary/records.py` (`forward_group` , `compute_group_loss` , `decode_group`)	`enable_records: true` , `event_records: true` , `candidate_pool: per_query`
Relational junction layer & column loss	`gliner2/models/boundary/records.py` (`_assign_logits` , `_column_loss`)	`record_link_mode: junction` , `record_link_column_weight: 0.3`
Multi-mention trigger ownership	`gliner2/models/boundary/records.py` (`compute_group_loss`)	`record_coreferent_ownership: true`
Learned trigger coreference & MLP distance	`gliner2/models/boundary/records.py` (`TriggerLink`)	`record_coref_link: true` , `coref_link_distance: mlp` , `coref_link_distance_dim: 64`
Validation-calibrated cluster merge	`gliner2/training/coref_calibration.py` , `tools/train/train.py`	`eval.coref_calibration` , `eval.coref_control`
Argument coreference & occurrence grouping	`gliner2/processing/argument_coref.py` , `gliner2/processor.py` , `gliner2/processing/boundary_preprocessing.py`	`training.expand_argument_coref: true`
Split positive/hard-negative role BCE & withhold unresolved	`gliner2/models/boundary/records.py` (`_list_field_bce` , `_unresolved_overlap_cols`), `matching.py`	`record_role_hard_negatives: 8` , `record_drop_anchor_field_loss: true`
Split positive/negative pair loss & absent-query cap	`gliner2/models/boundary/losses.py` (`candidate_pair_loss`), `model.py`	`pair_split_reduction: true` , `hard_negative_keep_all_when_absent: false`
Assignment-probability decode gate	`gliner2/inference/engine.py` (`_format_record_extractions`)	default in record formatting
Cluster & offset-gold evaluation metrics	`gliner2/training/eval_metrics.py` (`_gold_event_trigger_set` , `_gold_event_argument_set` , `event_cluster*`)	default in `compute_metrics`

9. References

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